

CoSA Data Aggregation Planning

Proof of Concept Findings

Prepared by AVP for CoSA | May 7, 2024

Summary

AVP developed two proofs of concept for two use cases proposed by CoSA to illustrate the challenges involved in building compelling arguments with data, from dataset identification to data visualization in order to assess the level and types of data support to provide members and other users.

AVP used Tableau Public, a free public version of the Tableau data visualization software, to generate the proofs of concept, linked below:

- [Use Case 1: Resources by State Dashboard](#)
- [Use Case 2: Budget per Capita](#)
- [Use Case 2: Budget per Capita Map](#)
- [Use Case 2: Budget as Percentage of Total Expenditures](#)
- [Use Case 2: Budget as Percentage of Total Expenditures Map](#)

Findings from the proofs of concept include:

- Contextual information on data sources and missing data will be necessary to provide for accurate interpretation and visualization of data.
- CoSA will likely need to provide cleaned versions of datasets for users in order for data to be useful.
- Many data creation and data visualization tasks may prove too difficult for CoSA members with basic data manipulation skills. CoSA may wish to prioritize providing interactive visualizations for its members to use over preparing and publishing its data for download.

- For some use cases, the right visualization may help to highlight gaps in the data. However, some interactive features, such as filters, may hide these gaps and mislead users.

Introduction

The goal of this Proof of Concept was to demonstrate the process, tools, and level of effort needed for successful data aggregation and result in a sample visualization of the end product. By creating a proof of concept, we hoped to gain valuable information on possible challenges and a general level of effort for the implementation phase. It is also anticipated that this proof of concept will be useful for demonstrating the value of following through on implementation to funders and stakeholders in future phases.

Methodology

Process

AVP set out to mimic the process that a CoSA member might follow in conceptualizing a visualization that satisfies a specific use case and then finding, manipulating, and presenting the data in a data visualization tool.

These steps included:

1. Identify a use case
2. Identify data needed and determine the right visualization/interactive to support the use case
3. Find the data
4. Determine the best data structure needed for the visualization
5. Clean/reshape the data
6. Create the visualization

Research Questions

In planning and building the proof of concept, AVP and CoSA were interested in exploring the following questions and documenting our process as it relates to them. These insights will help in determining and prioritizing what immediate and long-term solutions CoSA should pursue.

1. How did we decide what visualization would be effective?
2. How did we find the right data for the visualization?

3. What contextual information was necessary to understand the data?
4. How did we have to reshape or normalize the data for our purpose?
5. What data was missing that we had to supply ourselves?
6. How did we represent missing data that could not be filled in?
7. How did our reshaping of the data potentially change how it may be interpreted?
8. How difficult was it to combine different datasets?
9. What are some of the dangers in combining datasets, either from CoSA or from external sources? How could the reshaping and/or joining of the data be misleading?
10. What tools and skills were needed to create the visualization?
11. What challenges came up in creating the visualization? How did we handle missing data?
12. What contextual information is needed to understand the visualization?
13. How could the visualization be misleading?
14. What is gained from the visualized data that may be difficult to glean from the charts or tabular data in the CoSA report?
15. How accessible is the visualization?

Use Cases

Use Case 1

An archivist needs to write a new digital preservation policy and is looking for states with a similar budget, number of staff, and a higher DPCMM score that have written and shared policies.

Use Case 2

A state archivist wants to make a case for a higher budget by comparing the percentage archives budget of total state budget and per capita budget of states with similar agency placement.

Dataset identification

For both use cases we decided to create a master data set of states, districts, and territories, their regions, program types, agency placements, and several other data points, such as current budget, staffing (FTE), and DPCMM scores in order to create an interactive visualization in which CoSA members could compare other states similar to theirs. Much of this data is available through the most recent ARM survey and the DPCMM survey. For states that did not complete the ARM survey, the region, program type, and agency placement data could be supplied from CoSA's internal airtable.

Use Case 1: Resources

For the first use case, CoSA supplied a spreadsheet of Resource Center records including metadata they had compiled related to the DPC Access Framework and tagged with related States, where applicable.

Use Case 2: Population and Expenditure data

For this use case, we looked to the Census Bureau website for data on 2020 population and 2022 state expenditures. Population data for all states, districts, and territories is available through the ["2020 Census 118th Congressional District Summary File \(CD118\)"](#) and accessed through the Census Bureau's [data explorer portal](#). Expenditure data (for states only) is available through the ["Annual Survey of State Government Finances."](#) The [Tables data](#) had the information needed in the simplest form.

Dataset creation

For the master dataset, we downloaded the raw survey data from Sogolytics as a CSV file and worked with it in Google Sheets to reduce it to only the necessary data. After realizing the dataset included duplicate entries for states and some gaps in data, we asked CoSA for support and were supplied with a cleaned version of the survey data without duplicates, an internal Airtable for filling in some gaps for agency and program, and spreadsheets containing DPCMM scores and regions. This data was downloaded as CSV for use in Tableau.

Use Case 1: Resources

Working with the spreadsheet supplied by CoSA, we needed to split multi-valued fields (FrameworkElements and State) into separate tables of single-valued fields along with the ResourceID field, so that they could be related in Tableau similar to a database join. This manipulation required the use of complex cross-sheet formulas. Once the data was in three sheets, the data was downloaded as an Excel workbook to be uploaded into Tableau.

Use Case 2: Expenditures and Population

Population data was downloaded from the Census website data explorer as CSV and then opened in Google Sheets and transposed from columns to rows. Population statistics for districts and territories were copied and pasted into the spreadsheet manually. The data was then downloaded as a CSV for upload to Tableau.

Expenditure data was downloaded as a CSV file from the Census website, opened in Google Sheets, reduced to only expenditure data, and then transposed from columns to rows. Values

were in units of thousands, so they were multiplied by 1000 to get the total expenditure value for each state. The data was then downloaded as a CSV for upload to Tableau.

Data visualization

All datasets were uploaded into the same Tableau workspace and then related as follows:

- The main Resources table was related to both Framework Elements table and States table from the same workbook by Resource ID.
- The Master Data table was related to the Resources table, Population table, and Expenditures table by State/District/Territory.

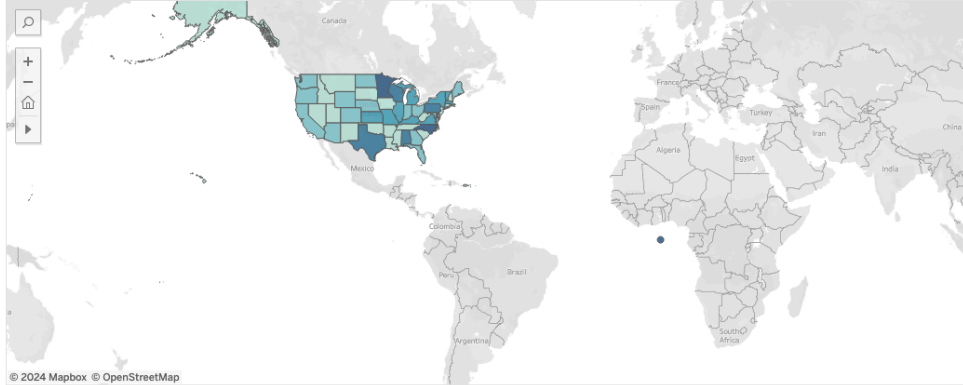
Use Case 1: Resources by State

To show number of related resources by state, we created a choropleth heat map and added filters for Region, Program, Agency, Budget, Staffing, Maturity (DPCMM Score), Year (of resource), and Framework Element, so that users could filter the results to find states similar to theirs and limit by year or framework element.

To show the actual resources associated with each state, we created a table grouped by state. Resources that were tagged with more than one state are repeated under each state. The same filters applied to the map were also applied to the table view.

Both visualizations were then combined in a dashboard so that filtering one visualization by selecting a state would also filter the other.

Related Resources by State/District/Territory



Year
(All)

Region
(All)

Agency Placement
(All)

Resource Type
(All)

Framework Element
(All)

Budget
\$400,000 \$14,134,795

Staffing (FTE)
4.00 73.00

Maturity (DPCMM Score)
0 43

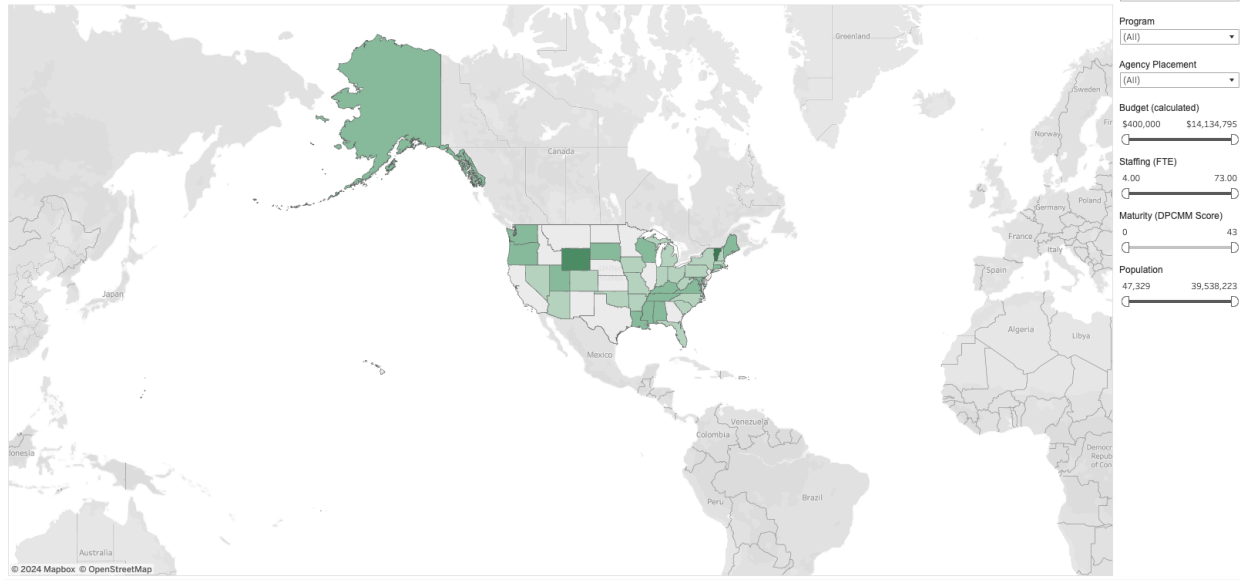
State / District ..	Resource Name		Publisher	Resource Type	Framework Elements	Year
Alabama	4C: Collaboration to Clarify the Costs of Curation		Null	Web Resource/Report	Designated Community	2014
	Case Study: Crowdsourcing the Alabama WWI Service Records		Null	Education/Training	Access, Open Standard / N..	2018
	Continuity of Operations Plan Template (AL)		Null	Form and Resource Sample	NA	
	Creating a Balance Among School, Work, and Life		Null	Web Resource/Report	Strategy	
	Crowdsourced Transcription Guide for Volunteers		Null	Education/Training	Access	2018
	Isabelle E Jarrott		Null	Education/Training	Access, Designated Comm..	2012
	Legal Admissibility of Public Records (AL)		Null	Policy/Guideline	NA	2004
	LOCKSS Program		Stanford Universit..	Tool	NA	2004
	Metadata Authority Description Schema (MADS)		Library of Congress	Standard	NA	2016
	Metadata Object Description Schema (MODS)		Library of Congress	Standard	Strategy	2018
	Records and Information Management (AL)		Null	Web Resource/Report	NA	
	Records Management Guide: Alabama Office of the Governor's Commu..		Null	Policy/Guideline	NA	2019

Dashboard view of heat map and table

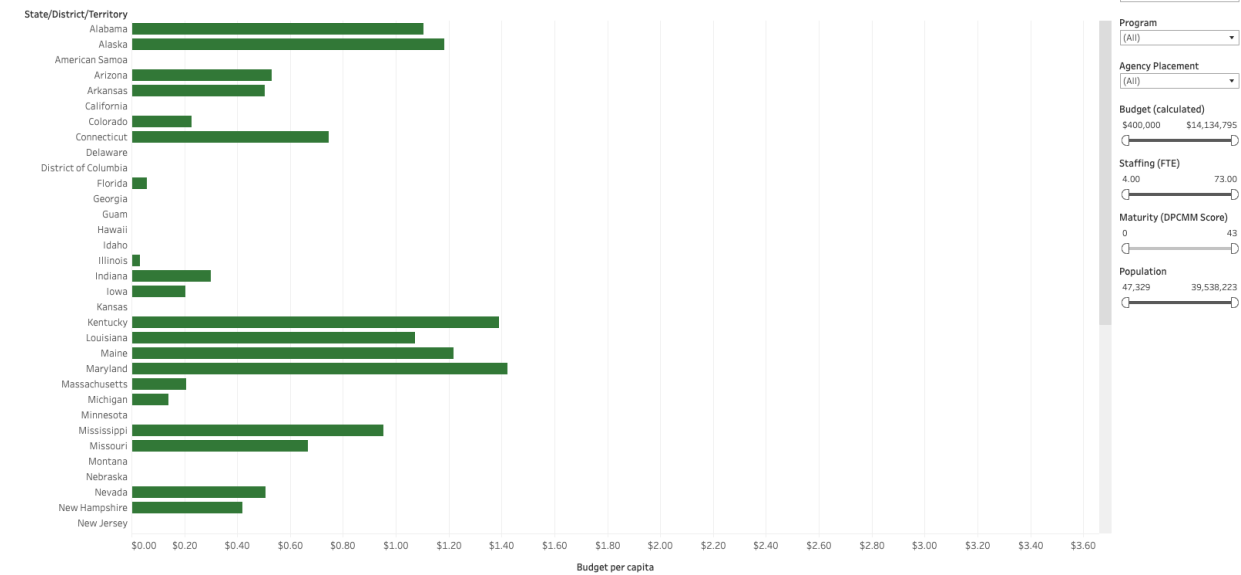
Use Case 2: Budget per capita

For this use case we created both a choropleth map and a bar chart to show archives budget per capita for each state/district/territory. We added filters for Region, Program, Agency, Budget, Staffing, Maturity (DPCMM Score), and Population, so users can compare their state to similar states.

Budget per Capita



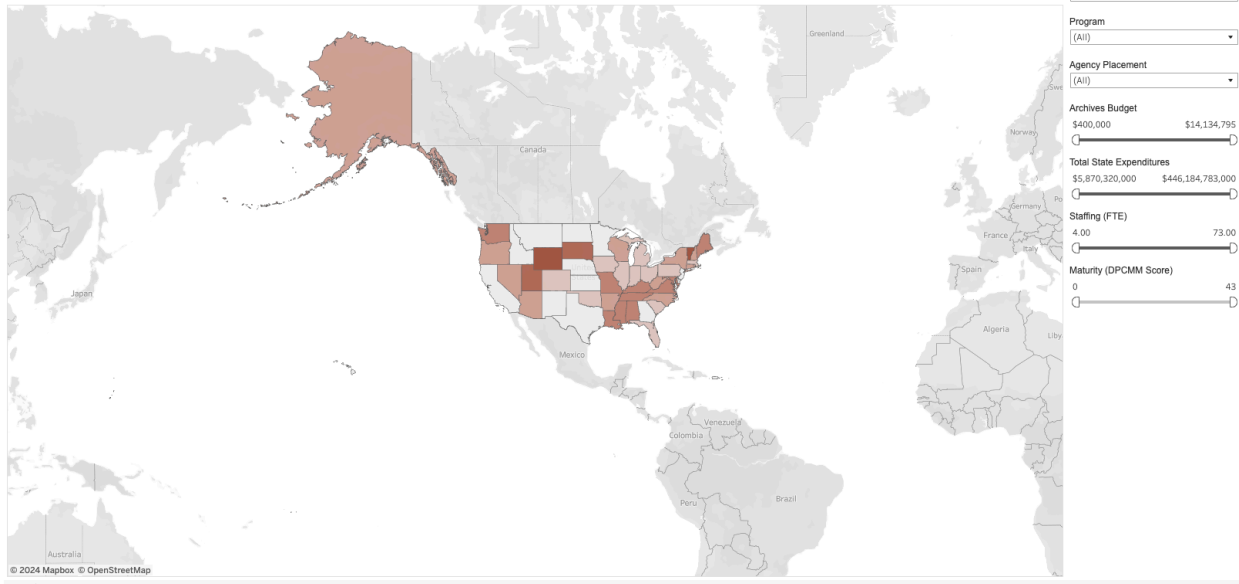
Budget per Capita



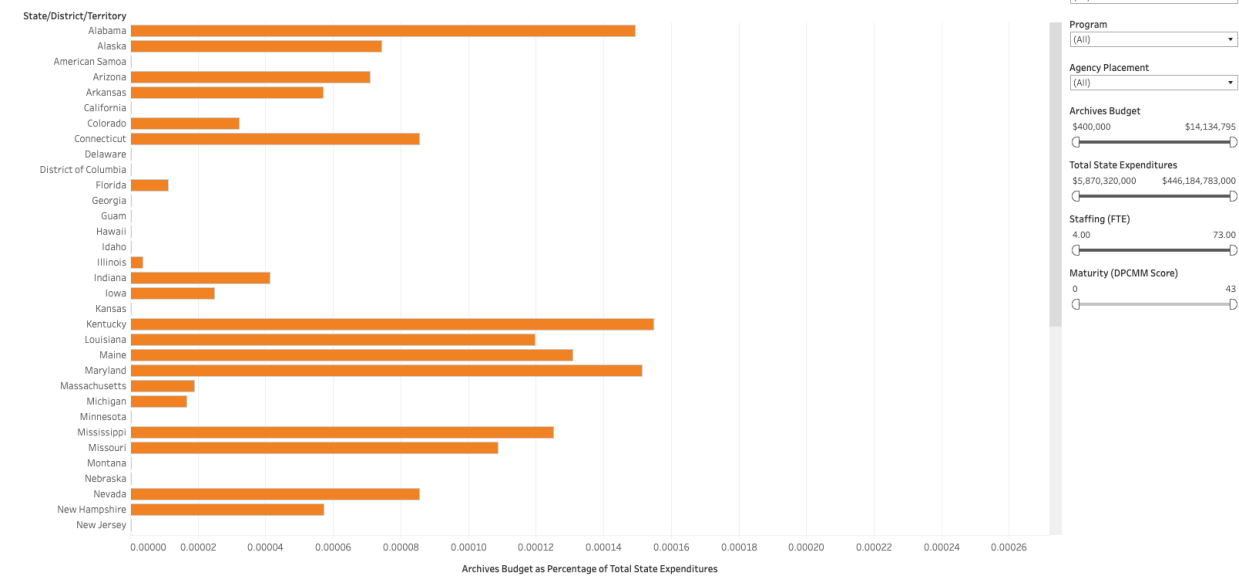
Use Case 2: Budget as percentage of total state expenditures

To show archives budgets as percentage of total state expenditures, we also chose a choropleth map and a bar chart and added the same filters but with Expenditures instead of Population.

Percentage of Total Expenditures 2022 Map



Percentage of Total Expenditures 2022



Details on the creation of these visualizations are included in the table below.

Process Findings

The following table documents the process AVP used to identify, find, acquire, clean, manipulate, and visualize the necessary data for the two use cases along with the estimated skill levels needed for each step. The last column includes findings on challenges and benefits that were

encountered (or could be encountered) during each step that may help inform CoSA's consideration of solution adoption.

	Summary	Experience / Familiarity Level	Challenges/Benefits
Dataset Identification			
Master data	Started with raw data from the 2022 ARM survey. Had to reference the survey instruments to understand which survey parts and columns were relevant. Some states had multiple responses. Asked CoSA for additional data to fill in gaps. CoSA supplied cleaned survey data and internal Airtable as well as internal spreadsheets mapping regions and DPCMM scores.	Intermediate - Requires some familiarity with CoSA surveys/reports as well as direct contact with CoSA staff for access to internal data.	- Context will be necessary (survey instrument or descriptive summary) to help users interpret ARM survey data and gaps in data. - Raw ARM survey data includes multiple responses for states. Consider making cleaned survey data available.
Use Case 1			
Resources data	Requested data from CoSA who shared internal data in spreadsheet form.	Basic - Data was readily available from CoSA.	CoSA would either need to make this data public or alert users that it is available upon request.
Use Case 2			
Population data	Searched the Census Bureau data explorer. State population data was immediately discoverable, but finding data for districts and territories required exact searches using Census Bureau terms (e.g. "United States Virgin Islands" instead of "Virgin Islands")	Intermediate - State population was easy to find, but keyword searching for data on districts and territories might lead a user to believe that it is not available unless they hit on the exact search terms.	CoSA could consider making this easier to find by making it available from CoSA in a normalized form, limited to the locations of interest.

	Summary	Experience / Familiarity Level	Challenges/Benefits
State expenditures data	A Google search easily identified the correct dataset. Some exploration of datasets was necessary to identify the best file to use. Additional searching was required to confirm that expenditure data for districts and territories is not available through the Census Bureau.	Basic – Some search and exploration may be necessary, but once the right file is identified, the data is self-explanatory. Somewhat more difficult to make sure that district and territory data is not available.	CoSA could consider making this easier to find by making it available from CoSA in a normalized form, limited to the locations of interest.
Dataset Creation/Normalization			
Master data	Copied/pasted relevant data from each data source into a Google Sheet. Manually checked values were correct after adding.	Basic – Once data was identified, basic copy/paste skills were necessary to compile the dataset. Sorting by state when pasting data in and manually adjusting values helped speed up the process, but this is not required.	Discovery of duplicate responses during the data creation process slowed progress. Making the cleaned survey data available will alleviate this issue.
Use Case 1			
Resources data	Two fields, "Framework Elements" and "State," are multi-valued, so it was necessary for Tableau visualization to create two additional sheets, one for each field, linking each to the Resources sheet by Resource ID, similar to a relational database. States were normalized to match values in the Master data sheet.	Expert – Knowledge of Google Sheet formulas was necessary to break out multi-valued fields into single-valued fields in new sheets.	When making data publicly accessible for download, CoSA will need to determine whether to clean and reshape the data as tables for easy upload to Tableau or other tools or to let the user do this work.

	Summary	Experience / Familiarity Level	Challenges/Benefits
Use Case 2			
Population data	Data was downloaded through the Census Bureau explorer as CSV, uploaded to Google Sheets and then transposed from columns to rows. District and Territory data was copied and pasted manually.	Basic - Download functionality was intuitive. Knowledge of transpose functionality in Google Sheets was required.	Considering the effort involved in retrieving and shaping the data, this dataset (as well as the expenditures data) would be a good candidate for reshaping and making available through CoSA's resource center for easy combination with CoSA datasets.
State expenditures data	Data was downloaded as a CSV file from the website. Unnecessary data was removed and the expenditure data was transposed from columns to rows. Values were originally represented in units of 1000, so the values were multiplied by 1000 to get the full value.	Basic - Knowledge of transpose and multiplication functions in Google Sheets was required.	
Data Visualization			
Use Case 1			
Use Case 1 - choropleth	Uploaded Master data and Resources workbooks to Tableau. Related Framework Elements and States sheets to Resources sheet on Resource ID field as relationships. Created	Intermediate - Requires knowledge of relating tables and creating calculated fields in Tableau.	Resources without states are all located on "Null Island" which is not immediately visible when centered on the U.S. The

	Summary	Experience / Familiarity Level	Challenges/Benefits
	choropleth heat map with density based on number of resources. Added filters for region, program, agency, year, framework element, budget, staffing, and maturity score.		choropleth map view also makes it difficult to see smaller states, districts and territories. Filters are misleading for states with no budget/staffing data.
Use Case 1 - table	Created a table list of resources grouped by state and added filters for region, program, agency, year, framework element, budget, staffing, and maturity score.	Intermediate - Requires knowledge of relating tables and creating calculated fields in Tableau.	Filters are misleading for states with no budget/staffing data.
Use Case 1 - dashboard	Combined choropleth and table with filters and used each as a filter for the other.	Intermediate - Requires knowledge of relating tables and creating calculated fields in Tableau.	- Same issues as above. - Filtering on null from the choropleth does not filter the table correctly.
Use Case 2			
Use Case 2 - percentage of expenditures bar chart	Uploaded Expenditures data to Tableau and related it to Master data by State/District/Territory. Created calculated field for archives budget as percentage of total expenditures. Created a bar chart for budget percentage by state. Added filters for region, program, agency, year, budget, staffing, maturity score, and	Intermediate - Requires knowledge of relating tables and creating calculated fields in Tableau.	The bar chart is effective at showing gaps in data, however, it is not immediately obvious if gaps are due to unreported archives budget data or if expenditure data (for districts and territories) is not available in the

	Summary	Experience / Familiarity Level	Challenges/Benefits
	expenditures. Added all information to the tooltip.		Census data. • Filters are misleading for states with no budget/staffing data.
Use Case 2 - percentage of expenditures choropleth	Plotted the above data as a choropleth heat map with density based on percentage.	Intermediate - Requires knowledge of relating tables and creating calculated fields in Tableau. Adjusting stepped colors to show missing budget/expenditure data is a challenging workaround.	• Displaying states with missing data as gray/white is a hacky workaround that may require manual adjustment when updating data for previous or subsequent years. • Same issues with filters as above
Use Case 2 - budget per capita bar chart	Uploaded Population data to Tableau and related it to Master data by State/District/Territory. Created calculated field for archives budget per capita. Created a bar chart for budget per capita by state. Added filters for region, program, agency, year, budget, staffing, maturity score, and population. Added all information to the tooltip.	Intermediate - Requires knowledge of relating tables and creating calculated fields in Tableau.	• The bar chart is effective at showing gaps in budget data. • Same issues with filters as above

	Summary	Experience / Familiarity Level	Challenges/Benefits
Use Case 2 - budget per capita choropleth	Plotted the above data as a choropleth heat map with density based on budget per capita.	Intermediate - Requires knowledge of relating tables and creating calculated fields in Tableau. Adjusting stepped colors to show missing budget data is a challenging workaround.	• Displaying states with missing data as gray/white is a hacky workaround that may require manual adjustment when updating data for previous or subsequent years. • Same issues with filters as above